

Quiz 1 (Version 4)

CAS CS 132: *Geometric Algorithms*

February 2, 2025

Name: Nathan Mull

BUID: 12345678

- ▷ You will have approximately 35 minutes to complete this exam.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.

1 Row Operations

Apply the row operations:

$$R_3 \leftarrow 5R_3$$

$$R_3 \leftarrow R_3 + 2R_2$$

$$R_2 \leftrightarrow R_3$$

from top to bottom to the following matrix. You must write down the intermediate matrices and row operations you used in your calculation.

$$\begin{bmatrix} 8 & 8 & 6 \\ -9 & -6 & -8 \\ -7 & 6 & -10 \end{bmatrix} \xrightarrow{R_3 \leftarrow 5R_3} \begin{bmatrix} 8 & 8 & 6 \\ -9 & -6 & -8 \\ -18 & -12 & -16 \\ -35 & 30 & -50 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 + 2R_2} \begin{bmatrix} 8 & 8 & 6 \\ -9 & -6 & -8 \\ -53 & 18 & -66 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3}$$

Solution.

$$\begin{bmatrix} 8 & 8 & 6 \\ -53 & 18 & -66 \\ -9 & -6 & -8 \end{bmatrix}$$

2 Reduced Echelon Forms

Determine the reduced echelon form of the following matrix. You must write down the intermediate matrices and row operations you used in your calculation.

$$\begin{bmatrix} -2 & -6 & 8 & 30 \\ 2 & 5 & -7 & -25 \\ -2 & -4 & 6 & 20 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 + R_1}$$

$$\begin{bmatrix} -2 & -6 & 8 & 30 \\ 0 & -1 & 1 & 5 \\ -2 & -4 & 6 & 20 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_1} \begin{bmatrix} -2 & -6 & 8 & 30 \\ 0 & -1 & 1 & 5 \\ 0 & 2 & -2 & -10 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 + 2R_2}$$

$$\begin{bmatrix} -2 & -6 & 8 & 30 \\ 0 & -1 & 1 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \leftarrow R_1 - 6R_2} \begin{bmatrix} -2 & 0 & 2 & 0 \\ 0 & -1 & 1 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \leftarrow -R_2}$$

$$\begin{bmatrix} -2 & 0 & 2 & 0 \\ 0 & 1 & -1 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \leftarrow -R_1/2}$$

Solution.

$$\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3 General Form Solutions

Determine a general form solution for a linear system whose augmented matrix is row equivalent to the following matrix (*Hint*. First determine if the following matrix is in reduced echelon form).

$$\begin{bmatrix} 1 & 0 & 3 & -5 & 0 & -4 \\ 0 & 1 & 5 & -5 & 1 & 4 \\ 0 & 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 3 & -5 & 0 & -4 \\ 0 & 1 & 5 & -5 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution.

$$x_1 = -4 - 3x_3 + 5x_4$$

$$x_2 = 8 - 5x_3 + 5x_4$$

x_3 is free

x_4 is free

$$x_5 = -4$$